YAKEEN NEET 2.0

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The p-Block Elements

Inorganic Chemistry | Lecture - 04

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Topics to be covered

Oxidation states and chemical Reactivity (15th, 16th, 17th)

1. Oxidation states & chemical Reactivity (15th, 16th, 17th)

2.

+3 more stable

- +3 oxidation state in case of arsenic, antimony and bismuth becomes increasingly stable with respect to disproportionation.

due to Inert pair effect

- Intermediate oxidations states have tendency to disproportionate.

Ex. :

Ex (i) +1 to +4 O.S of "N" have tendency to disproportionate into.

acidic medium

$$\overset{+3}{HNO_2} \longrightarrow \overset{+5}{HNO_3} + \overset{+2}{NO}$$

Intermediate O.S

Ex (ii) Intermediate O.S of "P" disproportionate into −3 & +5

$$\overset{+1}{H_3PO_2} \xrightarrow{\Delta} \overset{+5}{H_3PO_4} + \overset{-3}{PH_3}$$

$$\overset{+3}{H_3PO_3} \xrightarrow{\Delta} H_3PO_4 + PH_3$$

Q. Which of the following pair of elements in their +3 o.s is least stable w.r.t disproportionation.

(1) As, Sb

(2) Sb, Bi

(3) N, Bi

(4) N, P

- Nitrogen show all oxidation state in its oxides. (+1 to +5)

Ex. :

$\overset{+1}{N_2O}$ $\overset{+2}{NO}$ $\overset{+3}{N_2O_3}$ $\overset{+4}{N_2O_4}$ $\overset{+5}{N_2O_5}$

(i) Reactivity towards hydrogen

$E + H_2 \longrightarrow EH_3$ Hydrides

NH_3 PH_3 AsH_3 SbH_3 . .

e^- Rich Hydride (Lewis Base)

H-Bonding b/w molecules

High M.P & B.P

(ii) Reactivity towards oxygen

$$E + O_2 \longrightarrow \overset{+3}{E_2O_3} / \overset{+5}{E_2O_5}$$

PT

O.S↑, EN↑

Acidic↑ (O^{2-}/OH^-) $\overset{+5}{E_2O_5} > E_2O_3$ (Acidic)

Ex $N_2O_5 > N_2O_3$ (Acidic)

O^{2-} of N & P → Acidic

O^{2-} of As & Sb → Amphi.

O^{2-} of Bi → Basic

(iii) Reactivity towards halogens

$$E + X_2 \longrightarrow \overset{+3}{E}X_3 / \overset{+5}{E}X_5$$

$$\overset{+3}{E}X_3 < \overset{+5}{E}X_5 \text{ (cov. ch.)}$$

Ex

$SbCl_3 < SbCl_5$

$AsCl_3 < AsCl_5$

$PCl_3 < PCl_5$

** all Halides are Covalent Except BiF_3 → Ionic

N does not Form Pentahalides

$\cancel{NX_5}$

∵ due to absence of Vacant d

Max Covalency of N ⇒ 4

Amongst NX_3 only NF_3 is stable.

NCl_3, NBr_3, NI_3 Explosive

(iv) Reactivity towards metals

Electropositive.

Nitrogen + Metal → Metal Nitride (N^{3-})

$Mg + N_2 \rightarrow Mg_3N_2$

$Ca + N_2 \rightarrow Ca_3N_2$

Phosphorous + Metal → Metal phosphide (P^{3-})

$Ca + P_4 \rightarrow Ca_3P_2$

As + Metal → Metal Arsenide.

$Na^+ \quad As^{3-} \rightarrow Na_3As$

sod. Arsenide.

Sb + Metal → Metal Antimonide

Zinc Antimonide

$Zn^{2+} \quad Sb^{3-} \quad Zn_3Sb_2$

Oxidation state & trends in Chemical reactivity : Group 16th

- Common oxidation states → (Except oxygen) +4, +6 are more common.
- Tendency to show –2 decreases down the group. (T→B Met. ch ↑)

Reason :

Ex. :

- Oxygen show only –ve oxidation state as –2 except in OF_2.

+ve with Fluorine

Reason : EN of oxygen is high

EN of F > EN of O

O_2F_2 (+1) OF_2 (+2)

- Stability of +6 decreases while stability of +4 increases down the group.

Reason : Inert pair effect

- Bonding in +4 and +6 is primarily covalent.

(i) Reactivity with hydrogen

$$E + H_2 \longrightarrow H_2E$$

H_2O H_2S H_2Se H_2Te

e^- Rich Hydrides

→ H-Bonding ✓ (3d H-Bonding)

M.P & B.P ✓

(ii) Reactivity with oxygen

$E + O_2 \longrightarrow EO_2 / EO_3$

SO_2, SO_3, SeO_2, SeO_3

TeO_2 TeO_3 ... etc.

Both EO_2 / EO_3 Acidic

* SO_2 R.A $\rightarrow S^{+4} \rightarrow S^{+6}$ more stable
* TeO_2 O.A $\rightarrow Te^{+4} \rightarrow Te^{+2}$ more stable

(iii) Reactivity towards the halogens

$$E + X_2 \longrightarrow EX_2 / EX_4 / \overset{+6}{EX_6}$$

↳ EF_6 most stable

$EX_4 \Rightarrow$ See saw (sp^3d)

Ex. SF_4 AB_4L_1 l.p present at equatorial Position

SF_4	SeF_4	TeF_4
gas	liquid	Solid

M.W ↑se Attraction ↑ →

- All elements except oxygen form dichlorides and dibromides. These dihalides are formed by sp^3 hybridisation and thus, have tetrahedral structure.

Ex SF_2, SCl_2 → AB_2L_2 sp^3, Bent, e^- geometry

OCl_2 OBr_2 ✗ Chloride Bromide

- The well known monohalides are dimeric in nature.

EX — dimerisation

Cl_2O Br_2O ✓ oxides ✓

- Examples : S_2F_2, S_2Cl_2, S_2Br_2, Se_2Cl_2 and Se_2Br_2.

F_2O ✗ OF_2 ✓ Fluoride

- These dimeric halides undergo disproportionation as given below

$$\boxed{2\overset{+1}{Se_2}Cl_2 \rightarrow \underset{+4}{SeCl_4} + 3\underset{0}{Se}.}$$

e^- geometry tetrahedral

Molecular geometry or shape → Bent

More EN

Baad Mai likho

Baad Mai bolo

Suffix → ide

Oxidation state & trends in Chemical reactivity : Group 17th

F^-, Cl^-, Br^-, I^-

- All the halogens exhibit –1 oxidation state.
- However, chlorine, bromine and iodine exhibit + 1, + 3, + 5 and + 7 oxidation states also.

Chemical Bonding
Covalency Part
Homework

due to Presence of vacant d orbitals

Show +1, +3, +5, +7 O.S of Cl

- The higher oxidation states of chlorine, bromine and iodine are realised mainly with oxygen and fluorine.

* With oxygen → oxides ClO_2, ClO_3, BrO_3, I_2O_5 .. etc

* With Fluorine → Interhalogens
  ClF_3, BrF_3, BrF_5
  IF_7

Reason ⇒ High EN of O & F

Oxidizing nature of halogens :

Reason

$$X + e^- \rightarrow X^-$$

$ np^5 $ $\qquad$ $ np^6 $

Full Filled stable Configuration

$$\underbrace{F_2 > Cl_2}_{S.O.A} > \underbrace{Br_2 > I_2}_{W.O.A} \text{ (Oxidising Power)}$$

Halogens → Oxidising agent

Halides → ~~Oxidising agent~~

X^-

Reducing Property (Halides)

$$F^- < Cl^- < Br^- < I^- \text{ (Reducing Power)}$$

V. V. Imp

NCERT

Fluorine → strongest oxidising agent.

Reason (i) Low F–F Bond dissociation Enthalpy.
↳ (l.p–l.p Repulsions)

(ii) High Hydration Enthalpy of F^- (due to small size & high charge density)

$$F_2 \xrightarrow[\Delta H_{BDE}]{B.D.E} F \xrightarrow[\Delta H_{eg}]{E.G.E} F^-_{(g)} \xrightarrow[\Delta H_{hyd.}]{H_2O} F^-_{(aq)}$$

NOTE Halogen can oxidise halides of higher atomic no. in solution as well as solid phase

(oxidising power)

$F_2 > Cl_2 > Br_2 > I_2$

$$F_2 + X^- \longrightarrow F^- + X_2 \quad (X = Cl/Br/I)$$

$$Cl_2 + X^- \longrightarrow Cl^- + X_2 \quad (X = Br/I)$$

$$Br_2 + I^- \longrightarrow Br^- + I_2$$

$$\cancel{Cl_2 + F^- \longrightarrow Cl^- + F_2}$$

$$Cl_2 + NaBr \longrightarrow NaCl + Br_2 \checkmark$$

$$Cl_2 + NaI \longrightarrow NaCl + I_2 \checkmark$$

Reactions of Halogens with Water. (Oxidising Power of Halogens)

$F_2 > Cl_2 > Br_2 > I_2$

$$F_2 + H_2\overset{-2}{O} \longrightarrow H\overset{-1}{F} + \overset{0}{O_2}\uparrow$$

$$Cl_2 + H_2\overset{-2}{O} \longrightarrow \underset{\text{Hydrochloric acid}}{H\overset{-1}{Cl}} + \underset{\text{Hypochlorous acid}}{H\overset{-2}{O}\overset{+1}{Cl}}$$

$$Br_2 + H_2O \longrightarrow \underset{\text{Hydro Bromic acid}}{HBr} + \underset{\text{Hypo Bromous acid}}{HOBr}$$

$X = Cl/Br$

$$X_2 + H_2O \longrightarrow \underset{\text{Hydrohalic acid}}{HX} + \underset{\text{Hypohalous acid}}{HOX}$$

$$I_2 + \underbrace{H_2\overset{-2}{O} \longrightarrow HI}_{} + O_2\uparrow \quad (\text{Non spont Rxn})$$

Infact Reverse Rxn is spont.

Oxygen can oxidize Iodide in acidic medium.

$$\underline{\underline{I^-}} + \underset{\underline{\underline{O.A}}}{\overset{0}{O_2}} + H^+ \longrightarrow I_2 + H_2\overset{-2}{O}$$

Chemical Bonding

Bond dissociation Enthalpy order :-

(Bond strength)

$Cl_2 > Br_2 > F_2 > I_2$

Reason l.p - l.p Repulsion

B.L Kam Hai Par Jagda Jyada Hai (l.p - l.p)

B.L order

$F_2 < Cl_2 < Br_2 < I_2$

Normal. No Exception

Based on Size.

(i) Reactivity towards hydrogen

$E + H_2 \rightarrow HE$

Hydrides.

HF, HCl, HBr, HI

↓

H-Bonding ✓

M.P & B.P Higher than Expectation.

(ii) Reactivity towards oxygen

$E + O_2 \longrightarrow$ oxides (very less stable) (Except Fluorine)

↳ OF_2 O_2F_2 ~~Oxides~~

These are essentially Fluorides of oxygen.

OF_2 & O_2F_2 → more stable.

↓

Work as Fluorinating agents

Ex O_2F_2 is used to Recover Plutonium (Pu) From spent Nuclear Fuel in Form of PuF_6

Stability of O^{2-}

F > Cl > Br > I
EN

I >

Oxide of Iodine have Max. Ionic Character due to Low EN of Iodine

Extra strength ← $\overset{+\delta}{I} - \overset{-\delta}{O}$

$\Delta EN \uparrow$

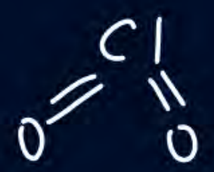
% I.C ↑

Cl >

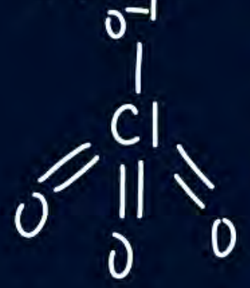
Size of Cl < Br & I

↑ Bond Formation Tendency with Oxygen is more

ClO_2



ClO_4^-



Br

EN of Br > EN of I

less Ionic ch in Bond with oxygen.

Br > Cl (size)

Multiple Bond Formation tendency less (weak Bonds)

Overall Bromine Lacks in Both conditions

thats why least stable oxide Formation.

Oxides of Chlorine

For Ex: Cl_2O ClO_2 Cl_2O_7 Cl_2O_6

- less stable & explosive in nature.
- oxidising agent. ✓

Oxides of Bromine :-

Ex Br_2O BrO_2 BrO_3

- least stable oxides amongst O^{2-} of Cl, Br, I
- oxidising agent.

Oxides of Iodine :-

Ex I_2O_4 I_2O_5 I_2O_7

- Insoluble solids ✓
- I_2O_5 O.A

Used in estimation of CO in Automobile industry.

Stability of O^{2-} : I > Cl > Br

(iii) Reactivity towards metals

Homework

Metal + Halogens $\longrightarrow$ Metal Halides

* Cov. ch. $MF < MCl < MBr < MI$ ($\because F^- < Cl^- < Br^- < I^-$)
Polarisability ↑
Covalent Ch. ↑
(Fajans Rule)

* Halides of higer O.S are more Covalent.

q^+ ↑
ϕ ↑
Covalent ch. ↑

$\overset{+3}{Sb}Cl_3 < \overset{+5}{Sb}Cl_5$

$SnCl_2 < SnCl_4$

$PbCl_2 < PbCl_4$

$UF_4 < UF_6$

(iv) Reactivity of halogens towards other halogens

Homework

Halogen + Halogen ⟶ InterHalogen.

X → Large size, less EN

X' → Small size, more EN

XX' → linear Ex- ClF

sp^3d XX'_3 → AB_3L_2 Bent T Ex- ClF_3, BrF_3

sp^3d^2 XX'_5 → AB_5L_1 Sq. Pyramidal Ex IF_5, BrF_5

sp^3d^3 XX'_7 → AB_7L_0 Pentagonal Bipyramidal. Ex. IF_7

Reactivity order ⇒ F I X Hai !! #MDJ

F_2 > Inter Halogens > Other Halogens $Cl_2/Br_2/I_2$

Reason: Low F-F B.D.E

Some Imp. Properties of group 15th, 16th & 17th Hydrides

Most Imp

Chemical Bonding

Done

15th	16th	17th
NH_3	H_2O	HF
PH_3	H_2S	HCl
AsH_3	H_2Se	HBr
SbH_3	H_2Te	HI

15th $E + H_2 \rightarrow EH_3$

16th $E + H_2 \rightarrow H_2E$

17th $E + H_2 \rightarrow HE$

Top to Bottom

E–H Bond length ↑se

→ Bond strength ↓se

→ Thermal stability ↓se

→ H^+ donation tendency in water ↑se

→ Acidic Nature ↑se ; pK_a values ↓se

→ Reducing Nature ↑se

(∵ B.L ↑se H Removal easy ⇓ Oxidation easy

Reducing Power ↑se

M.P → 1, 1, 2 (#MDJ)

B.P → 2, 1, 1 (#MDJ)

Explanation → Chemical Bonding

Bond Angle (15th & 16th)

Size of C.A ↑ Bond Angle ↓se

B.P (2, 1, 1) $\boxed{1 > 2 > 3 > 4}$

2 NH_3	1 H_2O	1 HF
4 PH_3	4 H_2S	4 HCl
3 AsH_3	3 H_2Se	3 HBr
1 SbH_3	2 H_2Te	2 HI

M.P 1, 1, 2

1 NH_3	1 H_2O	2 HF
4 PH_3	4 H_2S	4 HCl
3 AsH_3	3 H_2Se	3 HBr
2 SbH_3	2 H_2Te	1 HI

H-Bonding is Present in NH_3, H_2O & HF
M.P & B.P Higher than Expectation
in their Respective groups.

Chemical Bonding

Done

Interhalogens

Polar Bonds

Polar } Polar

dipole-dipole.

More attraction b/w Molecules ✓

More M.P & B.P

Halogens

Non polar Bonds.

N.P } N.P

L.D.F

Comparitively less attraction.

Low M.P. & B.P.

group-18th

Inert gases / Noble gases

✓ Except Rn & Og all are Found in atomosphere
(86) (118)

(Max Ar ~1% in dry air)

Noble gases

↳ Reason due to their Full Filled Configuration

ns^2np^6

Except He
$1s^2$

Chemical properties :-

* Very less reaction (Almost Unreactive)

(i) due ns^2np^6 configuration

(ii) due to Very High I.E (Na to denge)

&

+ve electron gain enthalpy (Na hi lenge)

* In 1962 first compound of Inertgas was prepared.

$Xe[PtF_6]$ Bartlett Compound.

Red colour Complex.

<u>Reason of Formation</u>:-

$Xe \approx O_2$ (I.E)

$O_2^+\{[PtF_6]^-$	$Xe^+\{[PtF_6]^-$
Red. Comp.	Red

After this Some more Compounds of Xe & Kr have been synthesised.

(only with High E.N., O & F)

Ex. XeF_6, XeF_4, XeO_2F_2, $XeOF_4$

KrF_2 etc.

"Their is no true compounds of He, Ne, Ar present"

Order of important properties for inert gases

He
Ne
Ar
Kr
Xe

(A) Size ↑se

(B) Surface area ↑se

(C) Ionization energy ↓se

(D) M.P. and B.P. ↑se (∵ L·D·F between atoms ↑se)

(E) Adsorption on charcoal ↑se (∵ surface area ↑se)

(F) Tendency of liquification ↑se (∵ L·D·F ↑se)

(G) Solubility in H_2O ↑se (∵ Dipole induced dipole Force ↑se)

(H) Non-ideal character ↑se (L·D·F)

Mass ↑se L·D·F ↑

Xe N·P } H_2O Polar

dipole induced dipole

Xenon Fluorides Compounds :-

Preparation

$$\underset{(\text{Excess})}{Xe} + F_2 \longrightarrow XeF_2$$

$$\underset{(1:5)}{Xe + F_2} \longrightarrow XeF_4$$

$$\underset{(1:20)}{Xe + F_2} \longrightarrow XeF_6$$

all are crystalline solids & fluorinating agents

XeF_6 can also prepared by Rxn of O_2F_2 with XeF_4

$$XeF_4 + O_2F_2 \rightarrow XeF_6$$

Hydrolysis of Xe Fluorides [Formation of xenon - oxygen compounds]

Redox Reactions

$$\overset{+2}{XeF_2} + H_2O \longrightarrow \overset{0}{Xe} + HF + O_2$$

$$\overset{+4}{XeF_4} + H_2O \longrightarrow \overset{+6}{XeO_3} + \overset{0}{Xe} + HF + O_2$$

(+2) + 4 = +6
$XeOF_4$
+4 + 2 = +6
XeO_2F_2
+6
XeO_3

$$\overset{+6}{XeF_6} \xrightarrow{\text{Partial } (1.H_2O)} XeOF_4 + HF$$

$$\overset{+6}{XeF_6} \xrightarrow{\text{Partial } (2\,H_2O)} XeO_2F_2 + HF$$

$$\overset{+6}{XeF_6} \xrightarrow{\text{Complete } (3H_2O)} XeO_3 + HF$$

(MDJ)

* Xe +6 Mai Rahegaa (Non Redox)
* HF Common
* Product Mai Xe Ke Sath Utna oxygen Jitna water liya.

F^- Exchange Reactions

$XeF_2 + PF_5 \rightarrow [XeF]^+ [PF_6]^-$

$XeF_4 + SbF_5 \rightarrow [XeF_3]^+ [SbF_6]^-$

$XeF_6 + CsF \rightarrow Cs^+ [XeF_7]^-$

$XeF_4 + RbF \rightarrow Rb^+ [XeF_5]^-$

$XeF_2 + BiF_5 \rightarrow [XeF]^+ [BiF_6]^-$

#(MDJ)

Xe Fluorides with p-Block → F^- donor

Xe Fluorides with s-Block → F^- acceptor

Comp?

Group 18

Uses Homework Notes

- All these gases except radon and oganesson occur in the atmosphere.
- Their atmospheric abundance in dry air is ~ 1% by volume of which argon is the major constituent.
- All the noble gases are monoatomic. They are colourless, odourless and tasteless. They are sparingly soluble in water.
- They have very low melting and boiling points because the only type of interatomic interaction in these elements is weak dispersion forces.
- Helium has an unusual property of diffusing through most commonly used laboratory materials such as rubber, glass or plastics.

- Helium is a non-inflammable and light gas. Hence, it is used in filling balloons for meteorological observations. It is also used in gas-cooled nuclear reactors. Liquid helium (b.p. 4.2 K) finds use as cryogenic agent for carrying out various experiments at low temperatures. It is used to produce and sustain powerful superconducting magnets which form an essential part of modern NMR spectrometers and Magnetic Resonance Imaging (MRI) systems for clinical diagnosis. It is used as a diluent for oxygen in modern diving apparatus because of its very low solubility in blood.

V. Imp

- Neon is used in discharge tubes and fluorescent bulbs for advertisement display purposes. Neon bulbs are used in botanical gardens and in green houses.

- Argon is used mainly to provide an inert atmosphere in high temperature metallurgical processes (arc welding of metals or alloys) and for filling electric bulbs. It is also used in the laboratory for handling substances that are air-sensitive. There are no significant uses of Xenon and Krypton. They are used in light bulbs designed for special purposes.

THANK YOU